Package 'irTools'

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Title R-tools for generating Utah's Integrated Report on water quality

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Description This package is a collection of R-tools developed by Utah Division of Water Quality to enhance and automate Utah's Integrated Report on water quality. The package includes tools for downloading data from EPA's Water Quality Portal and Attains, prepping raw data for WQ assessments, and WQ assessment and reporting tools.

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`applyScreenTable` *Apply screens to WQP data by selected table*

Description

Joins activity review inputs (detConditionTable, labNameActivityTable, activityMediaNameTable, masterSiteTable, paramTransTable, & activityCommentTable) to WQP data to apply decisions from input files to data.

Usage

```
applyScreenTable(
  data,
  wb,
  sheetname,
  flag_col_name,
  com_col_name,
  startRow = 1,
  na_dup_err = TRUE
)
```

Arguments

data	A merged WQP result object. Must include both narrowresult & activity files. May also be a post-fillMaskedValues() results object. Note: re-application of edited domain tables to an already screened dataset is not advised, and changes to the domain table likely will not be reflected in a re-screened dataset due to merging conventions.
wb	Full path and filename for Excel workbook containing screen table to be applied
sheetname	Name of sheet in workbook holding desired screening decisions
flag_col_name	Name to rename IR_FLAG column to.
com_col_name	Name to rename IR_COMMENT column to.
startRow	Row to start reading excel sheet from (in case additional headers have been added). Defaults to 1.
na_dup_err	Logical. If TRUE (default), exit function with error if IR_FLAG values are NA or if duplicates detected in combinations in the domain table for which InData=="Y". Set to FALSE to apply a screen table without checking for NA values in IR_FLAG.

Value

A data.frame object of WQP data with merged columns from input screening tables.

---

asmntDashboard *Run the assessment dashboard application Runs the assessment dashboard shiny app embedded in UT IR tools package.*

Description

Run the assessment dashboard application Runs the assessment dashboard shiny app embedded in UT IR tools package.

Usage

```
asmntDashboard()
```

assessEColi *Assess E.coli data at the year/site level.*

Description

Compares E.coli data to 30-day and max criterion standards using Scenarios A, B, and C, and assigns e. coli assessment categories to each site.

Usage

```
assessEColi(
  data,
  rec_season = TRUE,
  SeasonStartDate = "05-01",
  SeasonEndDate = "10-31"
)
```

Arguments

data	A prepped dataframe object (likely the ecoli object within the prepped_data list–prepped_data$ecoli) containing e.coli data at the site/use/parameter level with standards assigned.
rec_season	Logical. If TRUE, restricts assessments to recreation season data.
SeasonStartDate	A string in the form "mm-dd" to define beginning of rec season over which to perform assessments.
SeasonEndDate	A string in the form "mm-dd" to define end of rec season over which to perform assessments.

Value

Returns list with three objects: assessments from all Scenarios on all data, and ecoli assessments aggregated over scenario and year and rolled up to site level.

assessExcCounts *Perform assessments based on sample & exceedance counts (conventional & toxic assessments)*

Description

Performs site level (site-parameter-use-criterion specific) water quality assessments based on sample & exceedance counts by applying IR method flowcharts with user specified inputs.

Usage

```
assessExcCounts(
  data,
  min_n,
  max_exc_count = NA,
  max_exc_pct = NA,
  max_exc_count_id = NA,
  max_exc_pct_id = NA,
  id_cols = c()
)
```

Arguments

data	Site-use-parameter-criterion specific sample and exceedance counts(i.e. output from countExceedances())
min_n	Minimum sample size for sufficient data assessment (sample counts >= min_n are considered sufficient)
max_exc_count	Maximum allowable exceedance count for full support (exceedance counts > max_exc_count are considered not supporting) - one of max_exc_count or max_exc_pct must be specified
max_exc_pct	Maximum allowable exceedance percentage for full support (exceedance pcts > max_exc_pct are considered not supporting - one of max_exc_count or max_exc_pct must be specified
max_exc_count_id	Maximum allowable exceedance count for insufficient data with exceedances (for sites/parameters with insufficient data, exceedance counts > max_exc_count_id are considered insufficient data with exceedances) - one of max_exc_count_id or max_exc_pct_id must be specified
max_exc_pct_id	Maximum allowable exceedance percentage for insufficient data with exceedances (for sites/parameters with insufficient data, exceedance pcts > max_exc_count_id are considered insufficient data with exceedances) - one of max_exc_count_id or max_exc_pct_id must be specified

Details

When percentage arguments are supplied, specified percentages are used to calculate the maximum allowable number of whole samples exceeding a standard for that site/parameter to be considered supporting. Note that the min_n argument is >= and all other arguments are >.

Value

Returns input data frame with site-use-parameter-criterion specific assessment categories appended.

---

assessHFDO *Run HFDO assessments*

Description

Performs high frequency dissolved oxygen assessments per IR assessment methods. This includes checking for data sufficiency, calculating daily minima and averages, and 7-day/30-day moving averages.

Usage

```
assessHFDO(data, min_n = 10)
```

Arguments

data	HFDO data–for the time being is a test dataset built by EH using "hfdo_prep.R".
min_n	Minimum sample count for completing an assessment. This is used to generate the necessary number of consecutive days to calculate a moving average (averaging period length + min_n - 1).

Value

List object containing data used for minimum, 7-day, and 30-day assessments, full list of assessments by type, and a rollup to site-use assessments.

---

assessHFTemp *Run high-frequency temperature assessments*

Description

10 IR-2026 production method: individual critical-season readings versus the numeric max-temperature criterion (typically 20 C for 3A, 27 C for 3B/3C). An assessment unit is NS if any site exceeds `exceed_frac`.

Usage

```
assessHFTemp(
  data,
  value_col = "Water_Measurement",
  criteria_col = "Criteria",
  group_cols = NULL,
  exceed_frac = 0.1,
  start_month = 5,
  start_day = 15,
  end_month = 9,
  end_day = 30
)
```

Arguments

data	Accepted HF temperature records with a measurement column, a numeric `Criteria` column, timestamps, and site/AU identifiers.
value_col	Measurement column. Default `Water_Measurement`.
criteria_col	Numeric criterion column. Default `Criteria`.
group_cols	Optional site-level grouping columns. Default is IR site identifiers (not Sample_ID), matching IR-2026 production.
exceed_frac	Exceedance fraction for NS. Default 0.10.
start_month, start_day, end_month, end_day	Critical season (default May 15-September 30).

Details

This function assesses already accepted data. Interpolate, dry-sensor scoring, human review, and REJECT rules live in the highfreqQC repo (`prepHFTemp()`, `flagDryHFTemp()`, `applyHFTempRejectRules()`).

Value

List with `site_assessments` and `au_assessments`.

`assessLakeProfiles` *Run lake profile assessments*

Description

Performs lake profile assessments per IR assessment methods. This includes identifying stratified profiles & applying appropriate assessment methods for stratified & non stratified profiles.

Usage

```
assessLakeProfiles(
  data,
  do_crit = list(`3A` = 4, `3B` = 3),
  temp_crit = list(`3A` = 20, `3B` = 27),
  uses_assessed = c("3A", "3B")
)
```

Arguments

`data`	Lake profiles object returned by dataPrep step.
`do_crit`	List of beneficial use classes and associated dissolved oxygen criteria to use for assessment. Defaults to list("3A"=5, "3B"=3). This excludes chronic & ELS present criteria per assessment methods. Sites with site specific criteria will be assessed regardless of criteria specified in this argument. Objects in this list should match the uses_assessed argument.
`temp_crit`	List of beneficial use classes and associated water temperature criteria to use for assessment. Defaults to list("3A"=20, "3B"=27). This excludes chronic & ELS present criteria per assessment methods. Sites with site specific criteria will be assessed regardless of criteria specified in this argument. Objects in this list should match the uses_assessed argument.
`uses_assessed`	Vector of beneficial uses to be assessed for lake profiles. Defaults to 3A & 3B uses.

Value

Returns a list of lake profile assessment dataframes. profile_asmnts_mlid_param contains site/parameter level profile assessments, profile_asmnts_individual contains assessments for each individual profile, profile_criteria contains the criteria used for the profile assessment (excluding any site-specific criteria that may have occurred in the input dataset), profiles_long contains profile data in long format including the numeric criterion associated with each parameter, profiles_wide contains profile data cast to wide format.

`assignCriteria` *Assign numeric criteria to WQP data*

Description

Assigns general & site-specific numeric WQ criteria to WQP data. Does not calculate criteria w/ correction factors - this occurrs in dataPrep.

Usage

```
assignCriteria(
  data,
  crit_wb,
  crit_sheetname,
  ss_sheetname,
  crit_startRow = 1,
  ss_startRow = 1,
  rm_nocrit = TRUE,
  print = TRUE
)
```

Arguments

`data`	A merged, parameter translated, spatially referenced WQP result object. Must be post parameter translation step.
`crit_wb`	Full path and filename for workbook containing criteria.
`crit_sheetname`	Name of sheet in workbook holding criteria to be assigned.
`ss_sheetname`	Name of sheet in workbook holding site-specific criteria to be assigned. Should typically contain parameters corresponding to those in criteria table.
`crit_startRow`	Row to start reading criteria table excel sheet from (in case headers have been added). Defaults to 1.
`ss_startRow`	Row to start reading site specific criteria table excel sheet from (in case headers have been added). Defaults to 1.
`rm_nocrit`	Logical. If TRUE (default), remove records w/o numeric criteria before returning. If FALSE, function will pass through all records with flattened uses & any matching criteria - useful for passing through non-assessed parameters for other analyses.

Value

A flattened & expanded sample x use x parameter x criterion data.frame.

autoValidateWQPsites *Auto-validate USEPA WQP stations*

Description

Performs auto-validation on previously queried WQP stations combined with the existing master site list. Auto-validates any new sites (i.e. those not currently in the master site list) and appends them into existing master site list flagged for acceptance, rejection, or review. Checks for any new site types in new data. A warning message and a list of new site types is printed if new site types are encountered. Also re-auto-validates any sites in the master site list that have previously only undergone auto-validation (to account for any changes in auto validation process) and checks existing master site list for changes in AUs, selected site types, or property boundaries.

Usage

```
autoValidateWQPsites(sites_object, master_site_file, correct_longitude = FALSE)
```

Arguments

`sites_object`	Sites object queried from WQP to be reviewed.
`master_site_file`	Full path and filename of master site list as generated by autoValidateWQPsites function and manual site review application. Also contains a sheet of waterbody types and their associated IR_FLAGs (.xlsx).

Value

Exports a new, undated master site list to the location & filename provided by the user.

buildHFTempFeatures *Build daily dry-sensor features for HF water temperature*

Description

Aggregates high-frequency water temperature to one row per Sample_ID and Date with the continuous variability, air-water, rolling, and harmonic physics features used by the dry-sensor Random Forest. This is the single feature definition shared by model training (highfreqQC) and IR scoring (`flagDryHFTemp`); changing it requires bumping `hfTempFeatureVersion()` and retraining.

Usage

```
buildHFTempFeatures(hf_data, air_max_hour = 15)
```

Arguments

`hf_data`	data.frame with columns `DateTime_Local`, `Sample_ID`, `HF_Location_ID`, `Water_Measurement`, `tmmn_mean` (daily min air temp, C), and `tmmx_mean` (daily max air temp, C).
`air_max_hour`	Local hour assumed for the daily air temperature maximum when computing phase lag. Default 15.

Value

data.table with one row per Sample_ID and Date.

`composeExport` *Create dataset for secondary reviewer export from assessment dashboard*

Description

Pulls columns of interest from prepped data and calculates basic exceedance summaries for site-use-parameter assessments.

Usage

```
composeExport(
  prepped_data,
  toxics_assessed,
  conventionals_assessed,
  include_rejected = TRUE,
  rejected_data,
  create_workbooks = FALSE,
 params = c("Mercury", "pH", "Max. Temperature", "Arsenic", "Cadmium", "Chromium",
  "Aluminum", "Total Ammonia as N", "Calcium", "Minimum Dissolved Oxygen", "Hardness",
   "Magnesium", "Nitrate as N"),
  path
)
```

Arguments

`prepped_data`	A list of objects produced by dataPrep function, including toxics, conventionals, and ecoli dataframes.
`toxics_assessed`	A dataframe object of toxic assessments from the assessExcCounts function.
`conventionals_assessed`	A dataframe object of conventionals assessments from the assessExcCounts function.
`include_rejected`	A logical argument to specify whether the export workbooks should include rejected data or not. Defaults to true.
`rejected_data`	A dataframe object with all of the data screened out using the laboratory, media, detection-quantitation limit, parameter translation tables, and data prep screens. This is only required if include_rejected = TRUE.
`create_workbooks`	A logical argument to specify whether workbooks should be made of the specified parameters. Defaults to FALSE.
`params`	A vector of assessment parameter names to be placed in workbooks.
`path`	A folder path name in which to drop the exported workbooks. Only required if create_workbooks is true.

Value

A list composed of three dataframes: site-date-use-param records linked with aggregated daily values (if applicable), calculations, and exceedances, site-use-param summaries including sample and exceedance counts, and prepped E.coli data.

`countExceedances` *Calculate sample and exceedance counts (conventional & toxic assessments)*

Description

Compares water quality result values to standards to calculates sample and exceedance counts. This is geared towards conventional and toxic assessments.

Usage

```
countExceedances(
  data,
 group_vars = c("IR_MLID", "IR_MLNAME", "R317Descrp", "IR_Lat", "IR_Long", "ASSESS_ID",
  "AU_NAME", "BeneficialUse", "BEN_CLASS", "R3172ParameterName", "CriterionLabel",
    "SSC_MLID", "SSC_StartMon", "SSC_EndMon", "AsmntAggFun"),
  agg_exc = FALSE,
  agg_exc_as_n = TRUE
)
```

Arguments

`data`	A prepped water quality portal data object (i.e. output from dataPrep()). Must include IR_Value and ActivityStartDate.
`group_vars`	Vector of column names on which to group data when calculating sample counts and exceedances. This should not include any factors that prevent aggregation to site-scale assessments (e.g. date, time, etc.), but should include any columns that indicate a unique standard (e.g. season, acute v. chronic, etc.). See default for recommended.
`agg_exc`	Logical. If FALSE (default), individual samples are compared to the criterion as with non-aggregated criteria to count exceedances. If TRUE (default) aggregate samples by AsmntAggFun prior to counting exceedances.
`agg_exc_as_n`	Logical. If FALSE exceedance/support of aggregate water quality criteria (e.g. seasonal means) is indicated as 1 or 0 in ExcCount column. Only used if agg_exc==TRUE. If TRUE (default), set the ExcCount value of aggregate water quality criteria to equal the associated sample count. In this case, ExcCount==SampleCount indicates an exceedance and ExcCount==0 indicates no exceedance. This allows sample and exceedance counts for these types of criteria to pass through assessExc using the same input arguments.

Value

Returns sample and exceedance counts aggregated by grouping variables.

dataPrep *Final data prep step before assessments*

Description

Performs unit conversions & daily data aggregations. Also checks native vs. target activities and fractions, dissolved vs. total value checks, river/stream depth & flow checks, and generates value based data flags.

Usage

```
dataPrep(
  data,
  translation_wb,
  unit_sheetname = "unitConvTable",
  crit_wb,
  cf_formulas_sheetname,
  startRow_unit = 1,
  startRow_formulas = 1,
  split_agg_tds = TRUE
)
```

Arguments

data	A merged, translated, and numeric criteria assigned WQP results R-object. Target units for conversions are defined by units associated with assigned numeric critera.
translation_wb	Full path and filename for IR translation workbook (.xlsx).
unit_sheetname	Name of sheet in workbook holding IR unit conversion table. Defaults to "unitConvTable".
crit_wb	Full path and filename for workbook containing criteria.
cf_formulas_sheetname	Name of sheet in criterion workbook holding conversion factors and criterion formulas for criteria dependent on CFs.
startRow_unit	Row to start reading the unit conversion table excel sheet from (in case headers have been added). Defaults to 1.
startRow_formulas	Row to start reading the formulas table from (in case headers have been added). Defaults to 1.
split_agg_tds	Logical. If TRUE (default) split off TDS records w/ function assigned in AsmntAggFun into separate output. If FALSE, these records are passed through to conventionals output.

Value

A list of objects ready for assessments.

`fillMaskedValues` *Select quantitation limits, determine detection condition, and fill masked values as appropriate.*

Description

This function selects a single upper and lower quantitation limit for each record in result (if one exists in detquantlim), determines detection condition for each record based on the presence/absence of a result value, upper or lower quant limits, and whether the result value is above or below quant limits or <=0.

Usage

```
fillMaskedValues(
  results,
  detquantlim,
  translation_wb,
  detLimitTypeTable_sheetname = "detLimitTypeTable",
  unitConvTable_sheetname = "unitConvTable",
  detLimitTypeTable_startRow = 2,
  unitConvTable_startRow = 1,
  unitConvTable_startCol = 1,
  lql_fac = 0.5,
  uql_fac = 1
)
```

Arguments

`results`	A WQP results (must include narrow result, merged, wide objects OK) R-object name.
`detquantlim`	A WQP detection/quantitation limit file R-object. Should be matching complement to WQP results input.
`translation_wb`	Full path and filename for IR translation workbook (.xlsx).
`detLimitTypeTable_sheetname`	Name of sheet in workbook holding detection limit type names and ranked prioritizations table. Defaults to "detLimitTypeTable".
`unitConvTable_sheetname`	Name of sheet in workbook holding unit conversion table. Defaults to "unitConvTable".
`detLimitTypeTable_startRow`	Row to start reading the detLimitTypeTable excel sheet from (in case headers have been added). Defaults to 2.
`unitConvTable_startRow`	Row to start reading the unitConvTable excel sheet from (in case headers have been added). Defaults to 1.
`unitConvTable_startCol`	Column to start reading the unitConvTable excel sheet from (in case headers have been added). Defaults to 1.
`lql_fac`	Numeric - factor by which to multiply lower quantitation limit type values when filling masked data or other non-detects (e.g. below lql values). Default = 0.5.

uql_fac	Numeric - factor by which to multiply upper quantitation limit type values when filling masked data or other over limit values. Default = 1.

Value

Returns a data frame with new columns of selected limits, filled values, units, and detection condition appended to results input. In IR_DetCond column, "ND"=non-detect, "OD"=over detection, "NRV"=no result value & no ranked detection limit, "DET"=detection.

---

flagDryHFTemp *Flag potential dry-sensor days in HF water temperature for review*

Description

Builds daily features (buildHFTempFeatures), scores them with one dry-sensor random forest, adds harmonic physics evidence, and assigns a review disposition to every day. Neither the model nor physics rejects data; days needing review are written to a review file (analogous to validateSites() output for siteValApp()). A random sample of low-score days is added as an audit so missed dry days can be measured.

Usage

```
flagDryHFTemp(
  data,
  air_temp,
  model,
  value_col = "QC_Measure_Val",
  review_file = NULL,
  audit_n = 25,
  seed = 2028
)
```

Arguments

data	Output of prepHFTemp().
air_temp	Daily air temperature by site with HF_Location_ID, Date (or date), and tmmn_mean/tmmx_mean (or tmmn_c/tmmx_c) in degrees C.
model	Output of loadDryModel(), or a path to a model file.
value_col	Water temperature column used to build features. Default "QC_Measure_Val" (interpolated working value).
review_file	Optional path for an .xlsx review file with sheets review and model.
audit_n	Number of randomly selected "Retain" days added to the review as "Audit - low score". Default 25.
seed	Random seed for the audit sample.

Details

Reviewers fill Review_Decision with one of "Confirmed Dry", "Confirmed Wet", or "Unsure". Confirmed Dry days are rejected by applyHFTempQC(), and confirmed decisions become candidate training labels for the next model retrain in highfreqQC.

Value

List with `daily` (every scored day), `review` (days needing review), and `manifest` (model metadata).

`hfTempFeatureVersion` *Dry-sensor feature contract*

Description

`hfTempFeatureVersion()` identifies the feature definition in `buildHFTempFeatures()`. Saved models record the version they were trained with, and `loadDryModel()` refuses models built on a different version. `hfTempPredictorCols()` lists the Random Forest predictors; `hfTempPhysicsCols()` lists physics evidence kept outside the Random Forest so physics-selected labels are not circular.

Usage

```
hfTempFeatureVersion()

hfTempPredictorCols()

hfTempPhysicsCols()
```

Value

Character vector.

`hfTempPhysicsFeatures` *Daily harmonic amplitude ratio and phase lag versus air temperature*

Description

Fits a 24-hour harmonic to each day of HF water temperature and compares it to a sinusoid between the daily air min and max (peaking at `air_max_hour`). Air-exposed sensors have amplitude ratios near 1 and phase lags near 0.

Usage

```
hfTempPhysicsFeatures(hf_data, air_max_hour = 15)
```

Arguments

`hf_data`	data.frame with `DateTime_Local`, `Sample_ID`, `Water_Measurement`, `tmmn_mean`, `tmmx_mean`.
`air_max_hour`	Local hour of the assumed daily air maximum. Default 15.

Value

data.table keyed by Sample_ID and Date with `amplitude_ratio`, `phase_lag_hours`, and supporting columns.

`lakeDashboard` *Run lake dashboard shiny app Runs lake dashboard shiny app embedded in UT IR tools package.*

Description

Run lake dashboard shiny app Runs lake dashboard shiny app embedded in UT IR tools package.

Usage

```
lakeDashboard()
```

`loadDryModel` *Load the dry-sensor random forest*

Description

Loads the dry-sensor random forest saved by the highfreqQC training workflow. The file must contain `rf_model`, one trained `randomForest`. A random forest is already an ensemble of trees, so review scores come from that single forest. Files that hold a `model_list` of several forests are retired artifacts and are refused.

Usage

```
loadDryModel(path)
```

Arguments

`path`	Path to the model `.RData` file.

Details

If the file also contains `model_manifest`, its feature version and predictors must match `hfTempFeatureVersion()` and `hfTempPredictorCols()`, so training and scoring cannot silently drift apart. Without a manifest, the model version is taken from the file name and review thresholds use the defaults in `flagDryHFTemp()`.

Value

List with `model` (one `randomForest`) and `manifest`.

`predictDryHFTemp` *Score daily dry-sensor probabilities with one random forest*

Description

Score daily dry-sensor probabilities with one random forest

Usage

```
predictDryHFTemp(features, model, threshold = 0.9)
```

Arguments

`features`	Daily feature table from `buildHFTempFeatures()`.
`model`	One trained `randomForest`, e.g. `loadDryModel(path)$model`.
`threshold`	Probability at or above which a day is predicted Dry. Default 0.90.

Value

data.table with `Sample_ID`, `Date`, `avg_prob` (the forest's Dry probability), `predictedState`, and `probValues`. Days missing any predictor are dropped.

`prepAttains` *Prepares rolled up AU-Use and AU-Use-Parameter assessments for ATTAINS upload*

Description

Prepares rolled up AU-Use and AU-Use-Parameter assessments for ATTAINS upload

Usage

```
prepAttains(
  current_assessment,
  trans_wb,
  pull_attains = TRUE,
  assess_path,
  param_path,
  last_reporting_cycle,
  current_reporting_cycle,
  filesave = TRUE
)
```

Arguments

current_assessment	A list of assessment output objects to be rolled up to the AU-use and (for cat 5 only) AU-use-parameter resolutions.
trans_wb	Excel workbook containing translation tables connecting irTools outputs to ATTAINS inputs. Currently resides in lookup tables repo.
pull_attains	Logical. If TRUE (default), function uses readAttains function from the irTools package to pull assessment data from the previous reporting cycle. If FALSE, param_path and assess_path are needed.
assess_path	File path to ATTAINS assessments.csv from previous reporting cycle.
param_path	File path to ATTAINS parameters.csv from previous reporting cycle.
last_reporting_cycle	Year of previous IR reporting cycle.
current_reporting_cycle	Year of current IR reporting cycle.
filesave	Logical. If TRUE (default), function saves AU-Use-Assessments and Category 5 AU-Use-Parameter Assessments to .csvs in a folder selected by the user.

Value

Returns a list consisting of two objects: AU-Use-Assessments for all AU's and Category 5 AU-Use-Parameter Assessments.

---

prepHFTemp *Prepare HF water temperature for QC and assessment*

Description

Applies the rule-based HF temperature QC steps: rebuilds each deployment's expected timestamp series at its mode sampling interval, linearly interpolates short gaps, and flags rate-of-change outliers with a z-score on sequential differences (stratified by winter vs. non-winter within each deployment).

Usage

```
prepHFTemp(
  data,
  max_gap_minutes = 30,
  z_threshold = 9.5,
  winter_months = c(11, 12, 1, 2)
)
```

Arguments

data	data.frame of HF water temperature with at least Sample_ID, HF_Location_ID, DateTime_Local, and Water_Measurement. Columns that are constant within a Sample_ID (e.g. site attributes) are carried onto rebuilt timestamps.

`max_gap_minutes`
: Maximum time between the observations bracketing a gap for it to be interpolated. Default 30.

`z_threshold`
: Absolute z-score on sequential differences above which a record is flagged `is_outlier`. Default 9.5.

`winter_months`
: Months z-scored separately from the rest of the year. Default November-February.

Value

data.table with the input columns plus `Mode_Time_Difference_Minutes`, `gap_minutes`, `QC_Measure_Val` (interpolated working value), `Interpolation_Flag`, `Water_Measurement_Diff`, `winter`, `z_score`, and `is_outlier`. `Water_Measurement` is left unchanged (NA on rebuilt timestamps).

`prepWQPData` *Prep WQP Data*

Description

This function reads dataframes downloaded from EPA's WQP, removes duplicates, and converts result values and detection limit values with non-numeric characters to NA (with the exception of those with <|>|,). Converts all non-result value columns in nr, activity, detquantlim, and sites objects to character, including dates. Replaces hard coding in bookdown and readWQPFiles function.

Usage

```
prepWQPData(irdata, char_check = TRUE)
```

Arguments

`irdata`
: A list of WQP dataframes containing sites, narrowresult, activity, and detquantlim objects.

`char_check`
: Logical. If TRUE, the function produces a .csv showing all the special characters changed or converted to NA.

Value

A list consisting of nr, activity, detquantlim, sites, filtered sites, and merged_results (nr combined with activity) dataframes with special characters converted to NA for the ResultMeasureValue column. Saves an rdata file named "irdata_prepped.Rdata" containing the six data WQP objects.

readWQPFiles *Read WQP Files*

Description

Creates list object containing data frames needed to run irTools. Removes duplicates and creates objects containing orphan records. Identifies non-numeric values in numeric columns and allows user to edit values and save to new object.

Usage

```
readWQPFiles(
  file_select,
  narrowresult_file,
  sites_file,
  activity_file,
  detquantlim_file,
  orph_check = TRUE
)
```

Arguments

file_select	Logical. If TRUE, navigation window will pop up for manual selection of files. If FALSE, function requires full path names to files.
narrowresult_file	Full path and filename of narrowresult file queried from WQP to be reviewed (.csv).
sites_file	Full path and filename of sites file queried from WQP to be reviewed (.csv).
activity_file	Full path and filename of activity file queried from WQP to be reviewed (.csv).
detquantlim_file	Full path and filename of detquantlim file queried from WQP to be reviewed (.csv).
orph_check	Logical. Specifies whether function should perform orphan check between narrowresult and sites, activity, and detquantlim. If TRUE, produces list object containing orphan rows.

Value

List containing dataframes needed to run irTools package, as well as orphan records and edited non-numeric data tables.

`rollUp` *Summarize assessments to different spatial resolutions*

Description

Summarizes and aggregates assessments at one spatial resolution to another spatial resolution. For example, a collection of assessments conducted at the site/use/parameter level can be rolled up to the AU/use/parameter level or to the AU/use/level.

Usage

```
rollUp(
  data,
  group_vars = c("ASSESS_ID", "BeneficialUse", "R3172ParameterName"),
  cat_var = "IR_Cat",
  expand_uses = TRUE,
  print = TRUE
)
```

Arguments

`data`	A list of assessment output objects to be rolled up to a different spatial resolution.
`group_vars`	Vector of column names on which to group data for assessment rollups. Defaults to aggregate by ASSESS_ID, BeneficialUse, and R3172ParameterName.
`cat_var`	Name of assessment category column.
`expand_uses`	Logical. If TRUE (default), uses are expanded in the output to include all uses associated with group_vars, including unassessed groups which are marked as 'NA' in output dataframe column AssessCat. If FALSE, only assessed groups are included in the output.
`print`	Logical. If TRUE (default) print summary table of applicable assessment categories & parameters.

Value

Returns dataframe with assessment categories for each AU/BenUse/R3172ParameterName.

`siteValApp` *Run site validation shiny app Runs site validation shiny app embedded in UT IR tools package.*

Description

Run site validation shiny app Runs site validation shiny app embedded in UT IR tools package.

Usage

```
siteValApp()
```

updateCommentTable

Identify New WQP comment combinations and append them into IR Parameter Translation Workbook for review (sheet name activityCommentTable)

Description

This function identifies new ResultLaboratoryCommentCode/ResultLaboratoryCommentText/ActivityCommentText/Re combinations in WQP data for which an IR_FLAG has not yet been defined and appends them into the translation workbook for review.

Usage

```
updateCommentTable(
  data,
  translation_wb,
  sheetname = "activityCommentTable",
  startRow = 1,
  startCol = 1
)
```

Arguments

data	A WQP results (must include activity, merged objects OK) R-object or the full path and filename to a WQP result file (.csv). If input is a path/file name, datatype must be specified as "filepath".
translation_wb	Full path and filename for IR translation workbook (.xlsx).
sheetname	Name of sheet in workbook holding comment table. Defaults to "activityCommentTable".
startRow	Row to start reading the activityCommentTable excel sheet from (in case headers have been added). Defaults to 1.
startCol	Col to start writing the activityCommentTable excel sheet to (to the right of all formula based columns). Defaults to 1.

Value

Appends any new combinations in WQP data to translation_wb for review. This updates the input translation_wb with those new rows with system date in the "DateAdded" column.

updateDetCondLimTables

Update detection condition and limit type tables

Description

This function identifies new ResultDetectionConditionText & DetectionQuantitationLimitTypeName values in WQP data for which are not in IR translation tables and appends them into the translation workbook for review. New ResultDetectionConditionText & DetectionQuantitationLimitTypeName values generated by this function are derived from those in detquantlim that match (via merge) records in results input (this allows subsetting of these lists by other previously generated data screens if desired).

Usage

```
updateDetCondLimTables(
  results,
  detquantlim,
  translation_wb,
  detConditionTable_sheetname = "detConditionTable",
  detConditionTable_startRow = 2,
  detConditionTable_startCol = 1,
  detLimitTypeTable_sheetname = "detLimitTypeTable",
  detLimitTypeTable_startRow = 2,
  detLimitTypeTable_startCol = 1
)
```

Arguments

`results`	A WQP results (must include narrow result, merged, wide objects OK) R-object name.
`detquantlim`	A WQP detection/quantitation limit file R-object. Should be matching complement to WQP results input. New values derived from those that successfully merge to results input.
`translation_wb`	Full path and filename for IR translation workbook (.xlsx).
`detConditionTable_sheetname`	Name of sheet in workbook holding result detection condition names table. Defaults to "detConditionTable".
`detConditionTable_startRow`	Row to start reading the detConditionTable excel sheet from (in case headers have been added). Defaults to 2.
`detConditionTable_startCol`	Col to start writing the detConditionTable excel sheet to (to the right of all formula based columns). Defaults to 1.
`detLimitTypeTable_sheetname`	Name of sheet in workbook holding detection limit type names and ranked prioritizations table. Defaults to "detLimitTypeTable".
`detLimitTypeTable_startRow`	Row to start reading the detLimitTypeTable excel sheet from (in case headers have been added). Defaults to 2.
`detLimitTypeTable_startCol`	Col to start writing the detLimitTypeTable excel sheet to (to the right of all formula based columns). Defaults to 1.

Value

Appends any new values in WQP data to translation_wb for review. This updates the input translation_wb with those new rows.

`updateLabActMediaTables` *Update labNameActivityTable & activityMediaNameTable*

Description

This function identifies new LaboratoryName/ActivityTypeCode combinations & ActivityMediaName/ActivityMediaSubdivisionName combinations in WQP data for which an IR_FLAG has not yet been defined and appends them into the translation workbook for review.

Usage

```
updateLabActMediaTables(
  data,
  translation_wb,
  labNameActivityTable_sheetname = "labNameActivityTable",
  labNameActivityTable_startCol = 1,
  labNameActivityTable_startRow = 1,
  activityMediaNameTable_sheetname = "activityMediaNameTable",
  activityMediaNameTable_startCol = 1,
  activityMediaNameTable_startRow = 1
)
```

Arguments

`data`	A WQP results (must include activity, merged objects OK) R-object
`translation_wb`	Full path and filename for IR translation workbook (.xlsx).
`labNameActivityTable_sheetname`	Name of sheet in workbook holding lab name / activity table. Defaults to "labNameActivityTable".
`labNameActivityTable_startCol`	Col to start writing the labNameActivityTable excel sheet to (to the right of all formula based columns). Defaults to 3.
`labNameActivityTable_startRow`	Row to start reading the labNameActivityTable excel sheet from (in case headers have been added). Defaults to 1.
`activityMediaNameTable_sheetname`	Name of sheet in workbook holding activity / media/submedia name table. Defaults to "activityMediaNameTable".
`activityMediaNameTable_startCol`	Col to start writing the activityMediaNameTable excel sheet to (to the right of all formula based columns). Defaults to 3.
`activityMediaNameTable_startRow`	Row to start reading the activityMediaNameTable excel sheet from (in case headers have been added). Defaults to 1.

Value

Appends any new combinations in WQP data to translation_wb for review. This updates the input translation_wb with those new rows with system date in the "DateAdded" column.

updateMethodTrans *Update paramTransTable, paramFractionGroup, & WQPParamCASID tables in translation workbook*

Description

This function identifies parameter/fraction/unit/method combinations in WQP data for which an IR translation has not yet been defined and appends them to the translation table for review.

Usage

```
updateMethodTrans(
  data,
  translation_wb,
  methodsTransTable_sheetname = "paramMethodsTable"
)
```

Arguments

data	A WQP results (merged or narrow result) R-object
translation_wb	Full path and filename for IR translation workbook
methodsTransTable_sheetname	Name of sheet in workbook holding parameter translation table. Defaults to "paramTransTable".

Value

Appends any new parameter/fraction/unit combinations in WQP data to translation_wb. This updates the input translation_wb with those new rows with system date in the "DateAdded" column.

updateParamTrans *Update paramTransTable, paramFractionGroup, & WQPParamCASID tables in translation workbook*

Description

This function identifies parameter/fraction/unit/method combinations in WQP data for which an IR translation has not yet been defined and appends them to the translation table for review.

Usage

```
updateParamTrans(
  data,
  translation_wb,
  standards_wb,
  paramTransTable_sheetname = "paramTransTable",
  paramTransTable_startRow = 1,
  paramTransTable_startCol = 1,
  CASLookup_sheetname = "CASLookup",
```

```
  CASLookup_startRow = 2,
  CASLookup_startCol = 1,
  paramFractionGroup_sheetname = "paramFractionGroup",
  paramFractionGroup_startRow = 2,
  paramFractionGroup_startCol = 2
)
```

Arguments

data	A WQP results (merged or narrow result) R-object
translation_wb	Full path and filename for IR translation workbook
standards_wb	Full path and filename for IR standards workbook
paramTransTable_sheetname	Name of sheet in workbook holding parameter translation table. Defaults to "paramTransTable".
paramTransTable_startRow	Row to start reading the paramTransTable excel sheet from (in case additional headers have been added). Defaults to 1.
paramTransTable_startCol	Col to start writing the paramTransTable excel sheet to (to the right of all formula based columns). Defaults to 1.
CASLookup_sheetname	Name of sheet in workbook holding CAS lookup table from WQP. Defaults to "CASLookup".
CASLookup_startRow	Row to start reading the CASLookup excel sheet from (in case additional headers have been added). Defaults to 2.
CASLookup_startCol	Col to start writing the CASLookup excel sheet to (to the right of all formula based columns). Defaults to 1.
paramFractionGroup_sheetname	Name of sheet in workbook holding the WQP fraction names. Defaults to "paramFractionGroup".
paramFractionGroup_startRow	Row to start reading the paramFractionGroup excel sheet from (in case additional headers have been added). Defaults to 2.
paramFractionGroup_startCol	Col to start writing the paramFractionGroup excel sheet to (to the right of all formula based columns, note col w/ header 'spacer' in table). Defaults to 2.

Value

Appends any new parameter/fraction/unit combinations in WQP data to translation_wb. This updates the input translation_wb with those new rows with system date in the "DateAdded" column.

updateTable — *Update translation workbook sheets with new combinations found in the WQP dataset.*

Description

Joins translation workbook sheets (activityMediaNameTable, labNameActivityTable, detConditionTable, detLimitTypeTable, paramTransTable, paramFractionTable, unitConvTable) to WQP data to apply decisions from input files to data.

Usage

```
updateTable(data, trans_wb, sheet, cols, params = FALSE, standards_wb)
```

Arguments

data	A merged WQP result object. Must include both narrowresult & activity OR detquantlim files. May also be a post-fillMaskedValues() results object. Note: reapplication of edited domain tables to an already screened dataset is not advised, and changes to the domain table likely will not be reflected in a re-screened dataset due to merging conventions.
trans_wb	Full path and filename for Excel workbook containing screen table to be applied
sheet	Name of sheet in workbook holding desired screening decisions
cols	A vector of column names to be screened for new combinations of values to be placed in the translation workbook.
params	Logical. If true, will screen WQ parameters for CAS ID and match to CAS IDs in the standards workbook.
standards_wb	Full path and filename for Excel workbook containing WQ standards to be used to assess WQP data.

Value

Message reporting the number of new combinations found in the data that were added to the workbook sheet. Workbook sheet is updated in translation workbook, with blank IR_FLAG column for new combinations in need of review.

updateUnitConvTable — *Update IR unit conversion table*

Description

Updates IR unit conversion table based on combinations of native and target units as defined by numeric criteria tables.

Usage

```
updateUnitConvTable(
  data,
  translation_wb,
  sheetname = "unitConvTable",
  startRow = 1,
  startCol = 1
)
```

Arguments

data	A merged WQP results-activity R-object that has had detection limit values filled (passed through fillMaskedValues function) and numeric criteria assigned (passed through assignCriteria function).
translation_wb	Full path and filename for IR translation workbook (.xlsx).
sheetname	Name of sheet in workbook holding IR unit conversion table. Defaults to "unitConvTable".
startRow	Row to start reading the unit conversion excel sheet from (in case headers have been added). Defaults to 1.
startCol	Column to start reading the unit conversion excel sheet from (in case columns have been added). Defaults to 1.

Value

Updates unit conversion table in translation_wb with any new combinations of native and target units.

---

validateSites *Validate monitoring locations for the IR*

Description

Performs auto-validation on previously queried WQP stations combined with the existing master site list. Checks for any new site types in new data. A warning message and a list of new site types is printed if new site types are encountered.

Usage

```
validateSites(site_buffer = 10, sites, trans_wb, manual_path, slco = FALSE)
```

Arguments

sites	Sites object queried from WQP to be reviewed.
trans_wb	Full path and filename of translation workbook containing the waterbody type table used in th screening process.
manual_path	Path to workbook containing manual site changes from previous IR's
slco	Logical. Default is FALSE. If TRUE, function will use the grepl function to automatically merge SLCOWS and UTAHDWQ sites containing the same two-letter, 4-number combination code used to denote SLCo monitoring locations.

Value

Exports a new, undated master site list to the location & filename provided by the user.